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Phase 04: Integrated System Prototype – Court IQ

Overview

During Phase 04, we developed a working prototype (MVP) of our AI application, *Court IQ*, which predicts NBA player performance using machine learning. The system integrates real-world NBA data, trained models, and an interactive user interface to deliver predictions and insights.

Week 11: Application Development

In this phase, we built the core components of Court IQ. We developed the user interface using Streamlit (app.py), allowing users to select players and generate predictions.

We implemented backend logic in Python to load trained machine learning models using scripts such as:

- `model_loader.py` for loading saved models
- `predict.py` for generating predictions

We also integrated real-world data using:

- `fetch_player_gamelog_nba_api.py`, which retrieves NBA player statistics from an external API

This allowed the system to operate using live and historical data.

Week 12: System Integration and Testing

After building individual components, we integrated the full system by connecting the data pipeline, machine learning models, and user interface.

We tested the system by:

- Validating prediction outputs from predict.py
- Ensuring correct model loading through model_loader.py
- Verifying data accuracy from the NBA API script

We also resolved bugs, improved performance, and ensured the application produced consistent results across different players and game scenarios.

Week 13: Deployment Preparation

In the final phase, we prepared Court IQ for deployment. We organized the project structure and ensured all components worked together seamlessly.

We configured automation using:

- `.github/workflows/update_nba_data.yml` to update player data regularly

We finalized documentation and deployed the application using Streamlit, making it accessible as a fully functional AI system.

Result

The final result is a fully functional AI-powered sports analytics application that predicts NBA player performance using real-world data. Court IQ demonstrates the successful integration of data collection, machine learning, automation, and user interface design into a complete system.